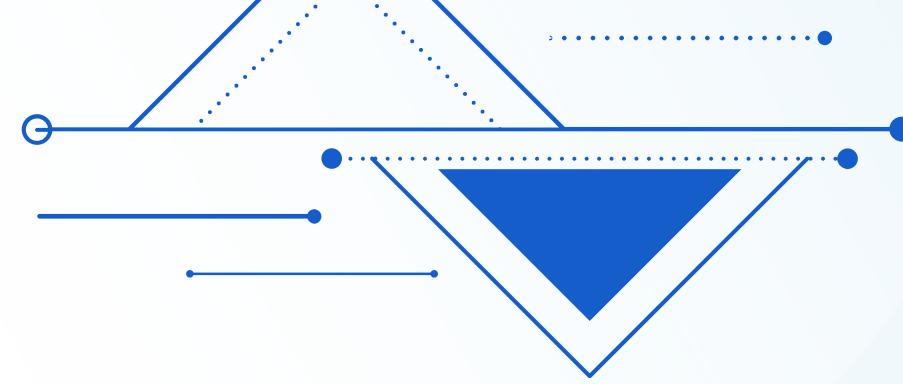


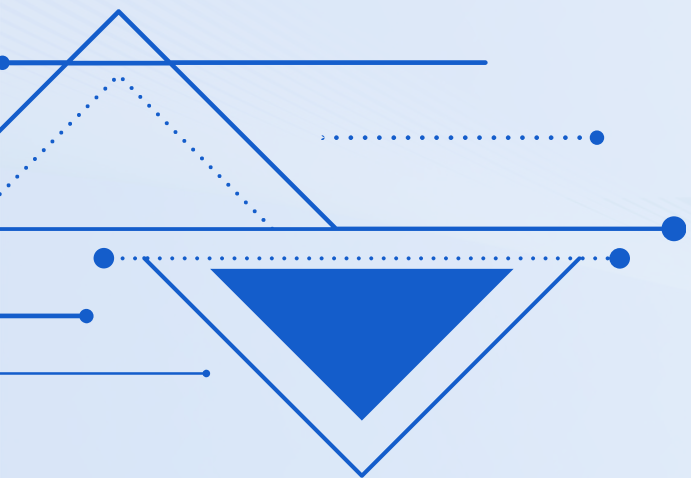
NTI



FPGA

UART (TX)

Presented by Mohamed Zanaty



UART:

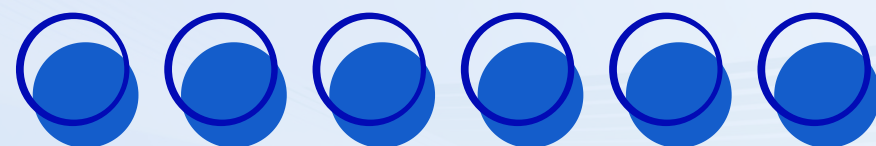
UART (Universal Asynchronous Receiver/Transmitter) is a serial communication protocol used to transfer data between digital systems without requiring a shared clock.

In this project, a UART Transmitter (TX) was designed using Verilog HDL. The transmitter takes parallel input data and converts it into a serial UART frame. The design includes an FSM controller, parallel-to-serial serializer, parity-bit generator, and MUX to control the transmission sequence.

The transmitter generates the UART frame in the following order:

Start Bit → Data Bits → Optional Parity Bit → Stop Bit

The design is parameterized, allowing the data width to be changed easily, and includes even/odd parity selection, busy status, and transmission completion detection.



what i made ?

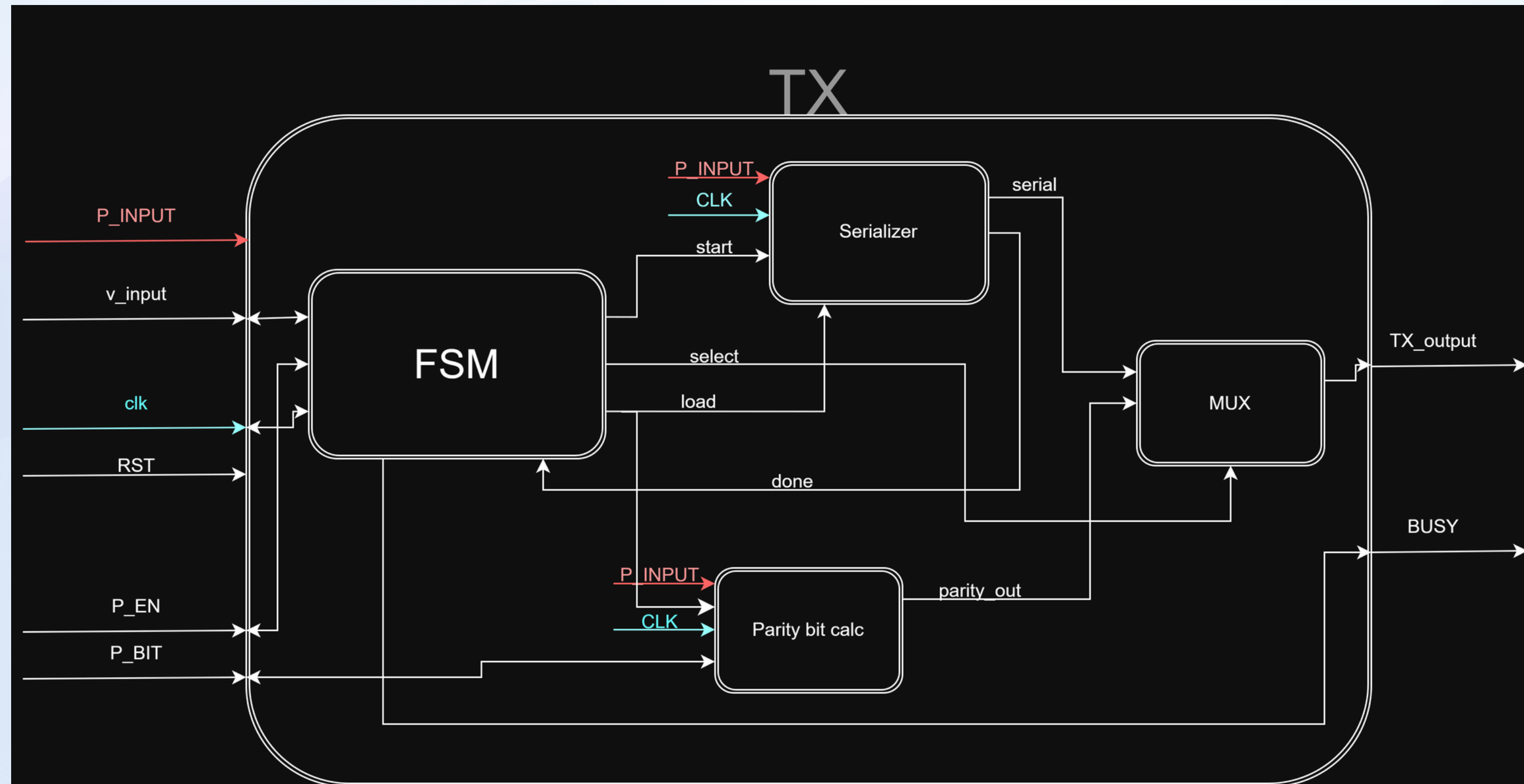


first, i made a flow chart to imagine what will be made.
i divided the project into 4 modules:

- FSM
- SERIALIZER
- PARITY_CALC
- MUX

The fsm controlles signals from others modules.
the serializer convert the parallel data to serial.
the parity_calc calculate the parity which is odd or even.
mux arrange the data and transmit to Tx_output.

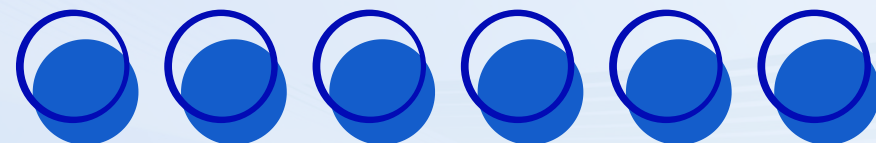
Flow Chart



FSM:

i choose 4 states for controlles:

- S0: that is the start to system and generate first bit (0) and make the system busy.
- S1: make mux get data from the serializer stage.
- S2: make mux get data from parity if P_en = (1).
- S3: that is stop and idle stage make the mux out (1).



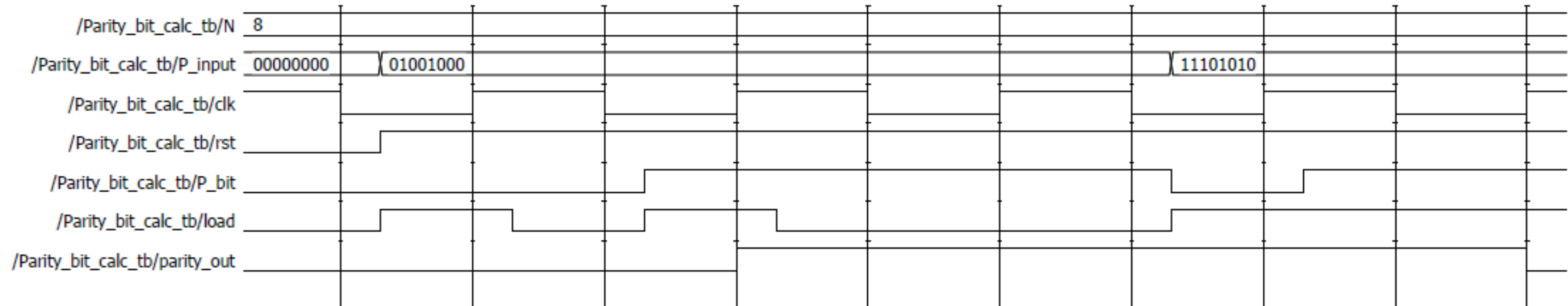
Parity_Calc:



this stage calculate the parity:

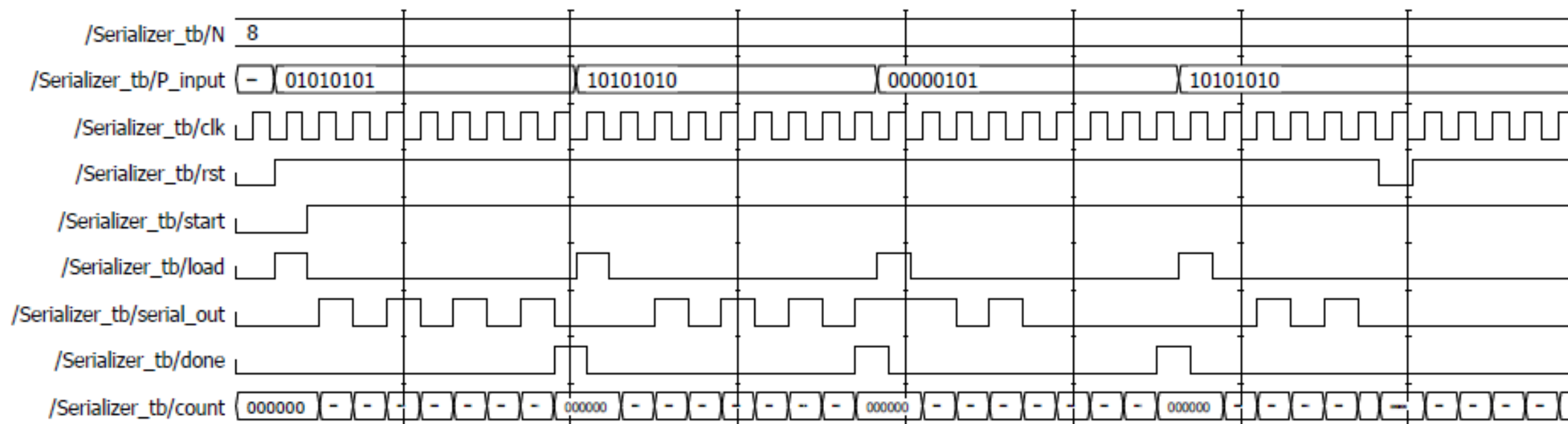
- if even the system xor the input data.
- if odd the system xnor the input data.

i tested it and works for my test cases (test even parity, test odd parity,)



Serializer:

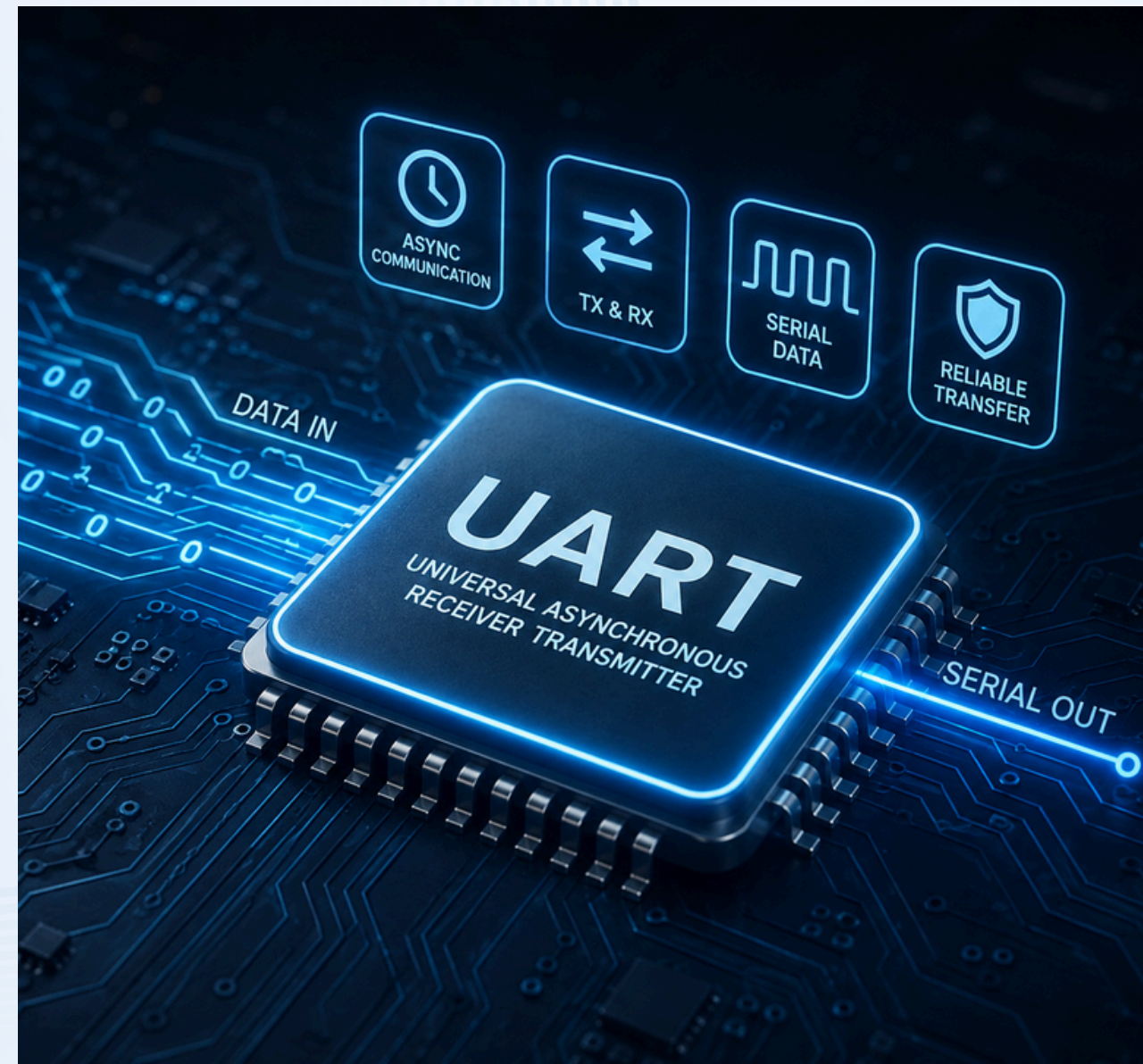
for serializer i made parallel to serial converter
it get data when load is (1)
and start when (1)
that is make a delay, if a new data come the serial will a delay and start but
i think it is okey because get a space for a start_bit (0).
i tested it and work well.



MUX:

Just a mux has 4 different staes and controlles the output:

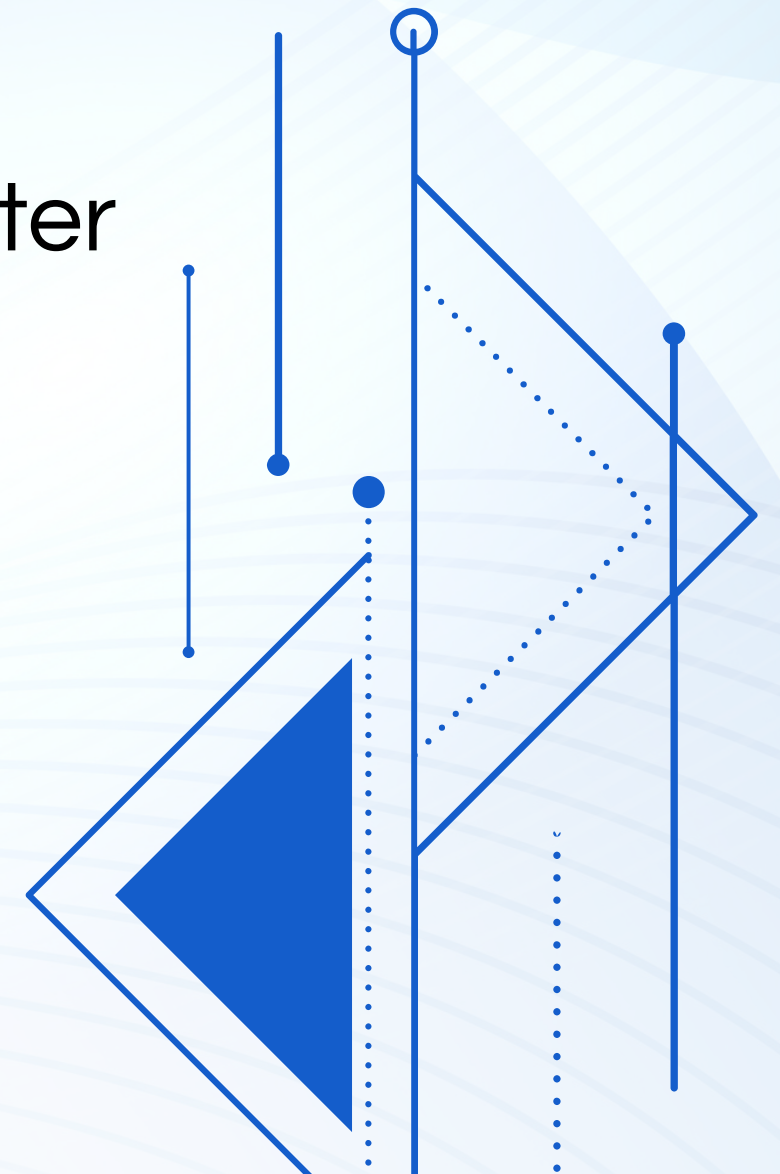
S0: out a start_bit (0).
S1: out the serial data.
S2: out the parity_bit.
S3:out the end_bit (0).



UART_TX:

for connect the 4 modules, i make it in structural model:
i tested it and works well but have some issues.

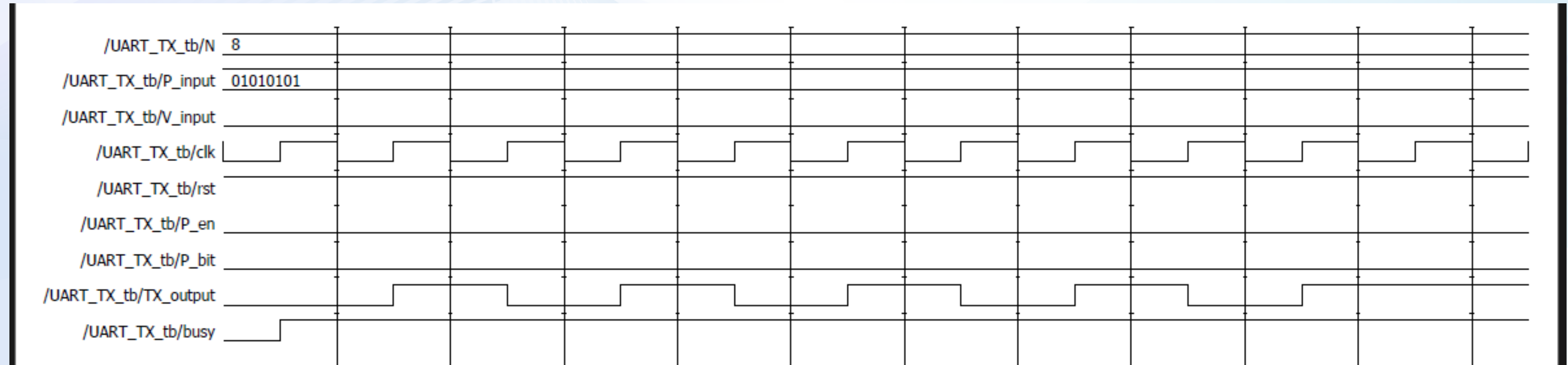
- first it delay one cycle if V_input is one and start the transmission.
- when insert input after first frame instantly (make v_in = 1 after finish the frame instantly), it make the output changes from 1(idle) to 0 but busy is high
- busy become high one more cycle.



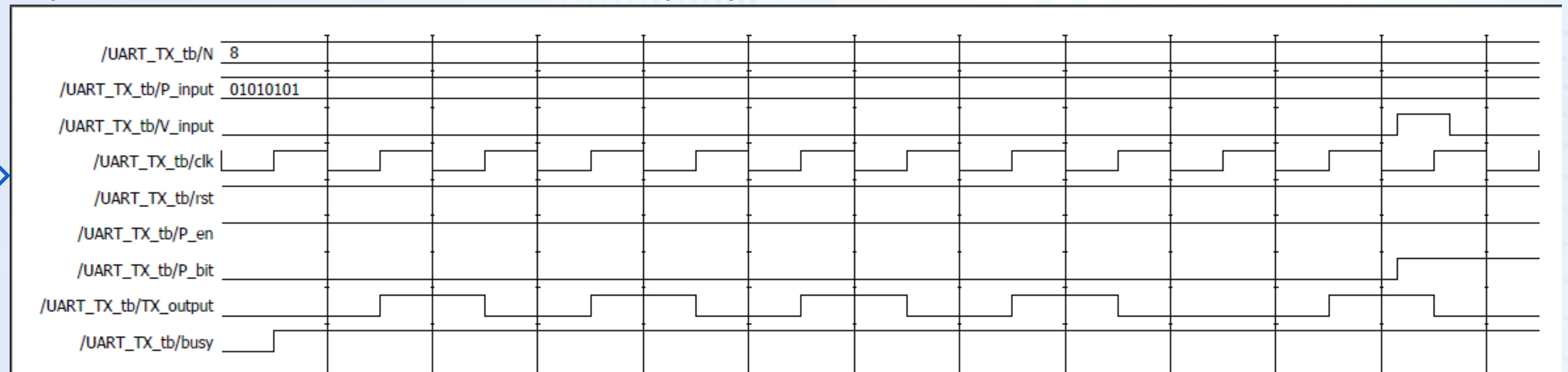
TEST CASES:



i test P_input = 01010101 and with out a prity.



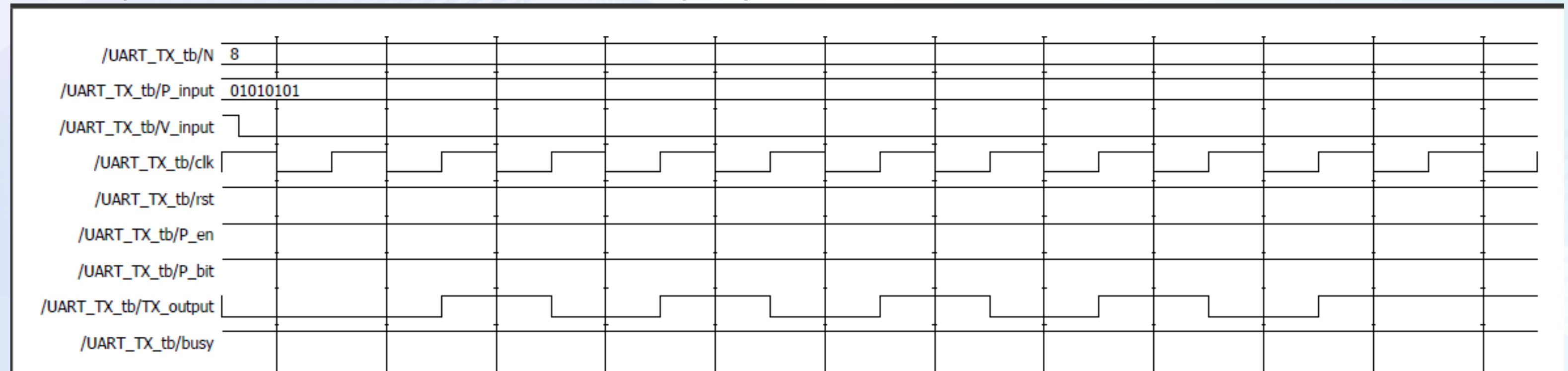
i test P_input = 01010101 and with an even prity.



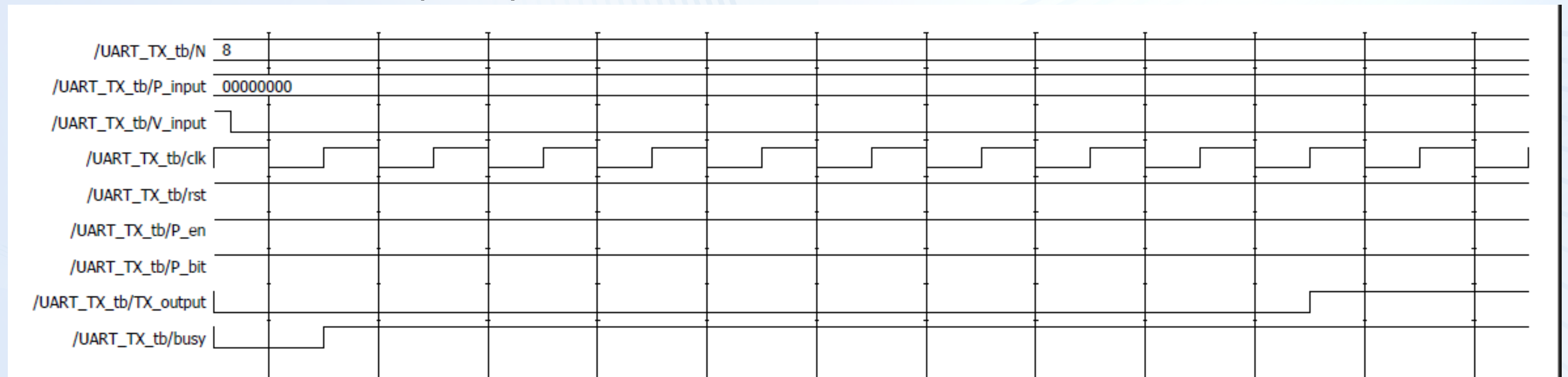
TEST CASES:



test P_input = 01010101 and with a odd prity.



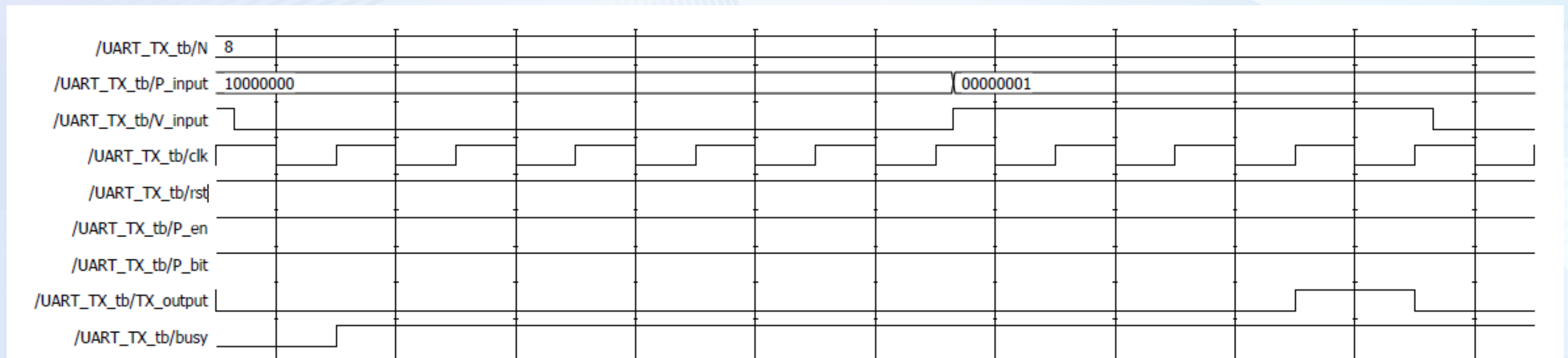
test P_input = 0 and with a odd parity.



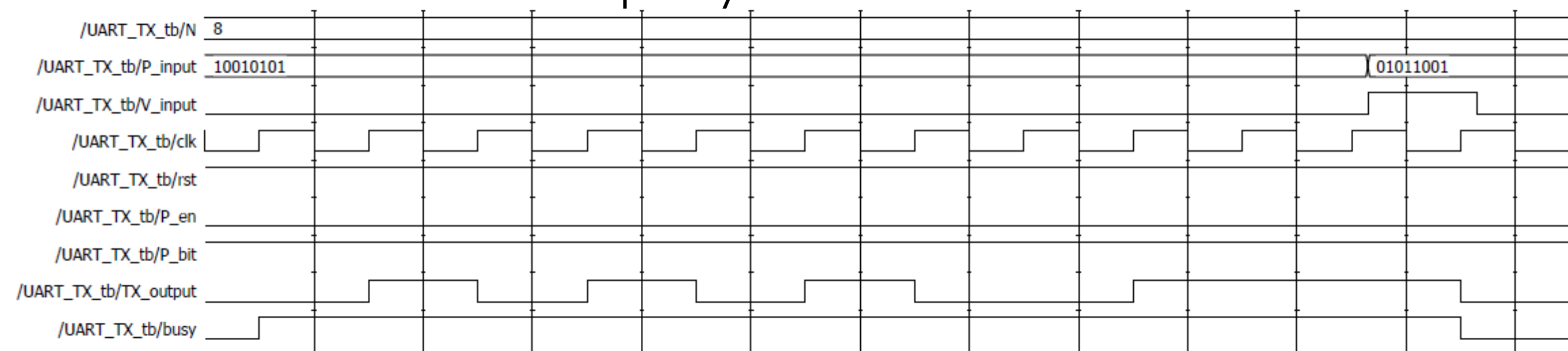
TEST CASES:



test P_input = 10000000 and with a odd prity and make V_in = 1 in the mid of transmtion. it works well but in end bit because i make V_in=1 it changes to 0



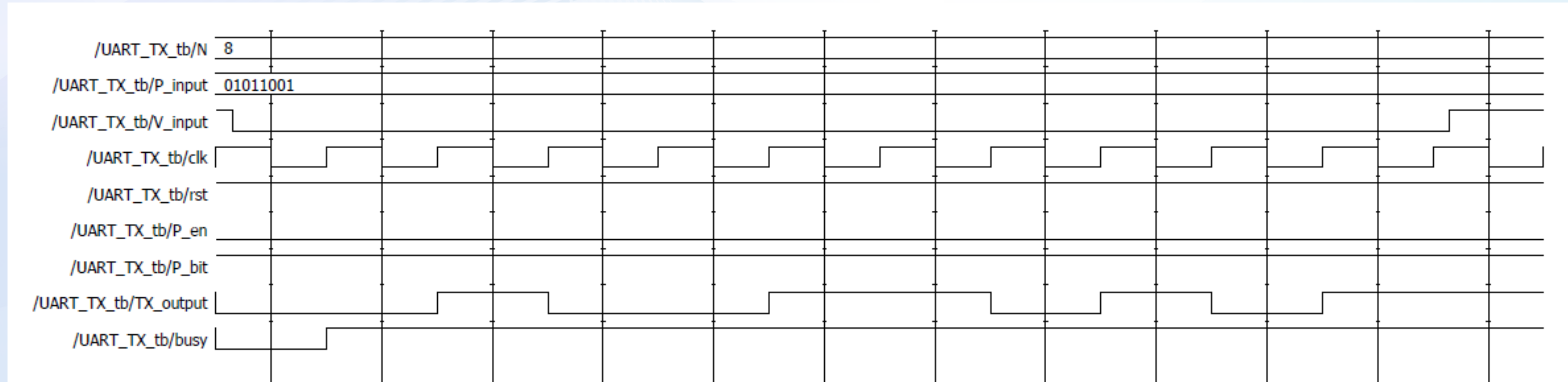
test P_input = 10010101 and with a odd parity.



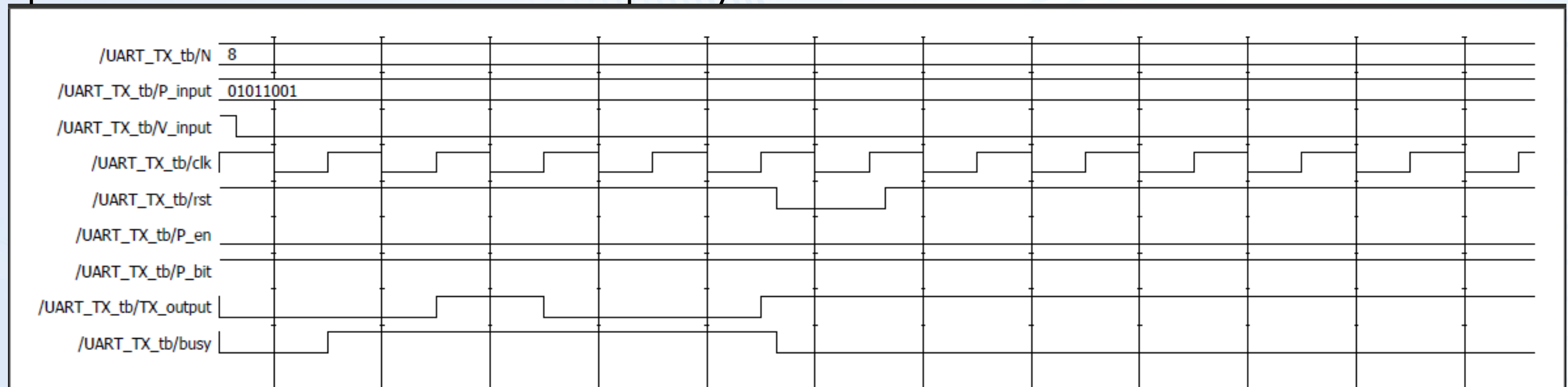
TEST CASES:



test P_input = 01011001 with odd parity



test P_input = 10010101 and with a odd parity and rst in mid.





THANK YOU

Presented by Mohamed Zanaty

NOTE: there is test bench files i uploded include the wave forms and transcript